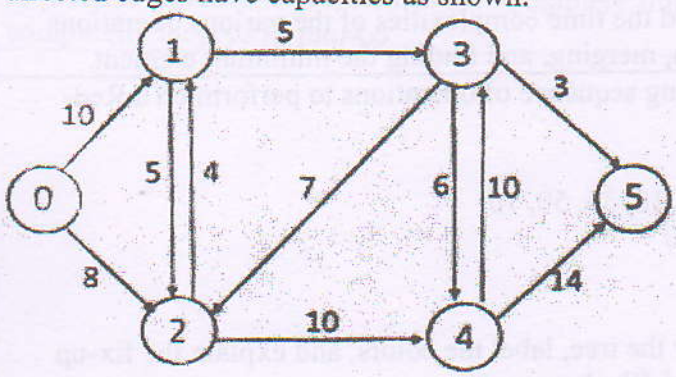
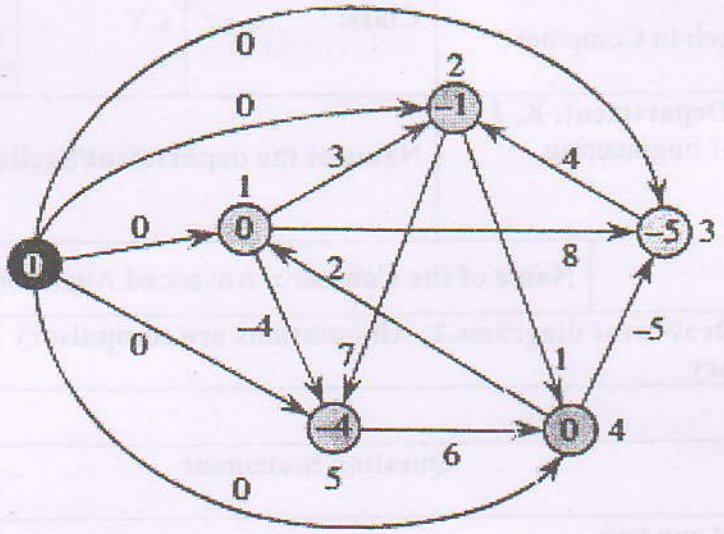
6/5/2025(M)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2025– April 2025			
Maximum Marks: 50	**Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code: 01 **Programme:**B.Tech in Computer Engineering	**Class:**	**LY**	**Semester:** VIII (SVU2020)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Center:** COMP		
Course Code: 116U01E852	**Name of the Course: :** Advanced Algorithms: Design and Analysis		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Explain the properties of Red-Black Trees. Describe how insertion and deletion operations maintain the balance of the tree	05
ii)	Explain the backtracking and branch-and-bound strategies. How do they differ in solving combinatorial problems such as the N-Queens problem and the Knapsack problem?	05
iii)	Explain the concept of the Master Method. Also, compute the complexity of the following recurrence relations using the master method. $T(n) = 9\,T(n/3) + n$ $T(n) = T(2n/3) + 1$	05
Q.2	Attempt any **one**	10
i)	Given the following flow network with source S and sink T, where the directed edges have capacities as shown:  a) Using the Ford-Fulkerson method, determine the maximum flow from source S to sink T. Show each augmenting path and flow updates. **(8 marks)** b) How does the time complexity of the Ford-Fulkerson algorithm differ	

	from that of Push-Relabel and Relabel-to-Front algorithms in computing maximum flow? **(2marks)**	
ii)	Apply Johnson's Algorithm to compute the shortest paths between all pairs of vertices. Show each step, including:  **a)** Explain the role of adding the extra vertex 0 in Johnson's Algorithm. Using the graph provided, describe how the Bellman-Ford algorithm is used to compute vertex potentials. ***(4 marks)*** **b)** Using the vertex potentials computed, perform edge reweighting to eliminate negative weights. Show the new edge weights. ***(3marks)*** **c)** After reweighting, outline how Dijkstra's algorithm is applied from each vertex. Why is Dijkstra's used at this stage instead of Bellman-Ford? ***(3 marks)***	
Q.3	Attempt any **one**	**10**
i)	Explain the operations of a Binomial Heap. Discuss its key properties, how it maintains its structure, and the time complexities of the various operations such as insertion, deletion, merging, and finding the minimum element.	
ii)	You are given the following sequence of operations to perform on a Red-Black Tree: • **Insert:** 40, 20, 60, 10, 30, 50, 70 • **Delete:** 20, then 40 Insertion **(5 Marks):** After each insertion, draw the tree, label the colors, and explain the fix-up case applied to restore Red-Black properties.	

	Deletion **(5 Marks):** After deleting nodes 20 and 40, show the updated tree, identify violations, and explain the fix-up steps taken to maintain the Red-Black properties.	
Q.4	Attempt the following	**10**
	a) Describe the Vertex Cover problem and explain why it is classified as NP-Complete. **(3 marks)** b) Develop a 2-approximation algorithm for the Vertex Cover problem. Present the steps of the algorithm clearly. **(3 marks)** c) Apply the algorithm to the following graph and show the resulting approximate vertex cover: Graph: Vertices = {A, B, C, D, E} Edges = {(A, B), (A, C), (B, D), (C, D), (C, E)} **(4 marks)**	
Q.5	Attempt the following	**10**
	a) Define the following complexity classes with appropriate examples: **(4 marks)** • P • NP • NP-Complete • NP-Hard b) Explain the concept of polynomial-time reduction. How is it used to prove that a problem belongs to the NP-Complete class? Illustrate your explanation with a suitable example. ***(4 marks)*** c) Why is proving a problem as NP-Complete important in computational complexity theory? ***(2 marks)***	